

WIFI_438

English User Guide

Reference build: WIFI_438_EN_FIX_LAN_DUALBAND_CASCADE.py

Kali Linux Wi-Fi / LAN analysis interface

WI-FI INTERFACE

Count: 0 **WIFI Router Detection**

DETECTED ROUTERS

Count: 0 **WIFI Router Detection**

BSSID	CHANNEL	SIGNAL	ESSID	BAND	CLIENTS	FRAMES	SECURITY

Scan clients associated with: --

ASSOCIATED WI-FI CLIENTS

Count: 0 **WIFI Router Detection**

STATIONS	DEVICE	VENDOR	SIGNAL	FRAMES	PROBES	SOURCE

LAN-SIDE DEVICES

Count: 0 **WIFI Router Detection**

VENDOR	ROLE	LIKELIHOOD	SC	MAC ADDRESS	SOURCE

HANDSHAKE

M1: --
M2: --
M3: --
M4: --
ABSENT

OBSERVED MAC ADDRESSES

MAC	VENDOR	CLASS	EVIDENCE	NOTES	SOURCE

BLOCK/CAPTURE DURATION (sec): 120

BLOCK

ROUTER DURATION (sec): 5

CLIENT DURATION (sec): 3

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POSSIBLE CAMERA DEVICES

Numero: 0

MAC	VENDOR	SCORE	LIKELIHOOD	TX/RX	SOURCE	DURATION

Actual main window of the English build.

Document date: 11 September 2026

This manual documents the English WIFI_438 build. Screenshots are captured from the actual program. The LAN and camera detail screenshots use illustrative values only so the window layout can be explained clearly.

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1. Purpose and scope

WIFI_438 is a Kali Linux desktop application for observing Wi-Fi networks, associated stations, Distribution System (DS) endpoints, possible LAN-side devices, handshake state, and traffic patterns compatible with camera devices. It combines live wireless scanning with passive capture analysis and keeps source/provenance information visible in the result tables.

The English build used for this guide includes the dual-band/cascade LAN correction: the presence of a related 2.4 GHz or 5 GHz radio no longer automatically removes a strongly supported DS-side LAN candidate. Direct observation of a MAC as a Wi-Fi station still has priority and prevents that MAC from being published as a LAN-side device.

Important interpretation rule: a LAN-side result means that the MAC behaves like an endpoint on the 802.11 Distribution System side. It does not, by itself, prove a physical Ethernet cable connection.

2. Safety and authorized use

Use the software only on networks and equipment that you own or for which you have explicit authorization. Passive observation can still expose device identifiers and network metadata. The BLOCK / DISRUPTION controls can transmit management traffic that may interrupt connectivity; they should be limited to controlled laboratory or authorized diagnostic use.

This guide explains what the controls do and how to interpret the interface. It does not provide procedures for disrupting third-party networks.

3. Requirements and startup

3.1 Core requirements

- Kali Linux or a compatible Linux desktop with Python 3 and Tkinter.
- An external USB Wi-Fi adapter whose driver supports monitor mode.
- Root privileges. The program checks for elevated privileges before monitor-mode operations.
- Required executables: iw, airodump-ng and tshark.
- Optional helpers used when available: nmcli, arp-scan, nmap, mergecap, Wireshark, pkexec/sudo and systemd-inhibit.

3.2 Starting the program

Keep the Python file in the folder where you want the program to create its working directories. A typical start command is:

```
sudo python3 WIFI_438_EN_FIX_LAN_DUALBAND_CASCADE.py
```

At startup, the program opens full screen. It creates a `wifi_audit_gui` working folder and a `CATTURE` capture archive next to the script. The selected external USB adapter is the only adapter that the monitor-mode activation routine attempts to reconfigure; the internal Wi-Fi interface is intentionally left untouched.

3.3 Keyboard and window controls

Control	Function
F11	Toggle full-screen mode.
Esc	Leave full-screen mode / close an in-app detail window when supported.
+	Increase GUI vertical size.
-	Reduce GUI vertical size.
EXIT	Close the application.
NIGHT VERSION / DAY VERSION	Switch between light and dark interface themes.

4. Main screen overview

WI-FI INTERFACE

REFRESH WI-FI INTERFACES ENABLE MONITOR MODE **SCAN WI-FI NETWORKS** NIGHT VERSION EXCLUSIVELY WITH WIFI CHIPSETS THAT CAN BE CONFIGURED IN MI- **EXPORT GRAPH** **EXIT**

DETECTED ROUTERS

Count: 0 **STOP SEARCH** WiFi Router Detection 0%

BSSID	CHANNEL	SIGNAL	ESSID	BAND	CLIENTS	FRAMES	SECURITY

SCAN CLIENTS OF SELECTED AP Scan clients associated with: --

ASSOCIATED WI-FI CLIENTS

Count: 0 **STOP SEARCH** 0%

STATIONS	DEVICE	VENDOR	SIGNAL	FRAMES	PROBES	SOURCE

LAN-SIDE DEVICES

Count: 0

VENDOR	ROLE	LIKELIHOOD	SC	MAC ADDRESS	SOURCE

BLOCK/CAPTURE 0%

BSSID: PASSIVE CAPTURE BSSID: --

CLIENT:

CHANNEL:

START **STOP** DURATION (sec): 120 **EXPO**

BLOCK

ROUTER:

START **STOP**

CLIENT:

DISRUPTION BSSID: -- / CLIENT: --

START **STOP** DURATION (sec): 5

☐ STOP WHEN HANDSHAKE IS FOUND PAUSE (sec): 3 **EXPORT H**

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HANDSHAKE

M1: --
M2: --
M3: --
M4: --

ABSENT

OBSERVED MAC ADDRESSES

MAC	VENDOR	CLASS	EVIDENCE	NOTES	SOURCE

POSSIBLE CAMERA DEVICES

Numero: 0

MAC	VENDOR	SCORE	LIKELIHOOD	TX/RX	SOURCE	DURATION

DETTAGLI **BLOCCA DISPOSITIVO SELEZIONATO**

Figure 1 - Main English interface. The layout is intentionally dense so scanning, capture and analysis remain visible together.

Area	Purpose
Top bar	Select the Wi-Fi interface, enable monitor mode, scan networks, switch theme, open debug, export the graph and exit.
Detected Routers	Shows BSSID, channel, signal, ESSID, band, client count, frame count and security.
Associated Wi-Fi Clients	Shows stations associated with the selected access point.
LAN-Side Devices	Lists DS-side endpoints that pass the LAN confidence logic.
Block/Capture	Contains passive capture and controlled disruption controls.
Handshake	Shows M1/M2/M3/M4 observation and the final handshake state.
Observed MAC Addresses	General classification table for MAC addresses seen during analysis.
Possible Camera Devices	Shows devices whose traffic pattern is compatible with a camera/always-on IoT profile.

5. Wi-Fi interface and monitor mode

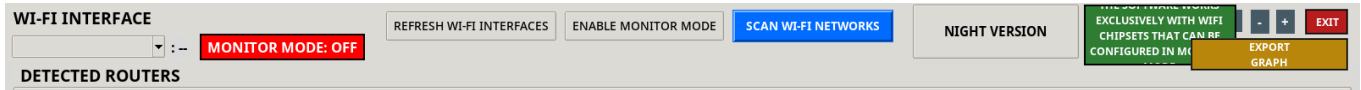


Figure 2 - Wi-Fi interface, monitor-mode controls and global buttons.

5.1 Selecting the adapter

1. Press REFRESH WI-FI INTERFACES to update the interface list.
2. Select the external USB Wi-Fi adapter in the drop-down list.
3. Check the origin indicator. The program distinguishes external USB interfaces from internal PC interfaces.
4. Press ENABLE MONITOR MODE. The status label changes when monitor mode is confirmed.

The activation routine verifies that the selected adapter is external USB, checks whether the associated PHY declares monitor-mode support, temporarily disconnects only that adapter from NetworkManager, changes it to monitor type, and verifies the result. If activation fails, the code attempts to restore that USB interface to managed mode.

If MONITOR MODE remains OFF, do not start a wireless scan. Verify the USB adapter, driver support and root privileges first.

6. Detecting routers

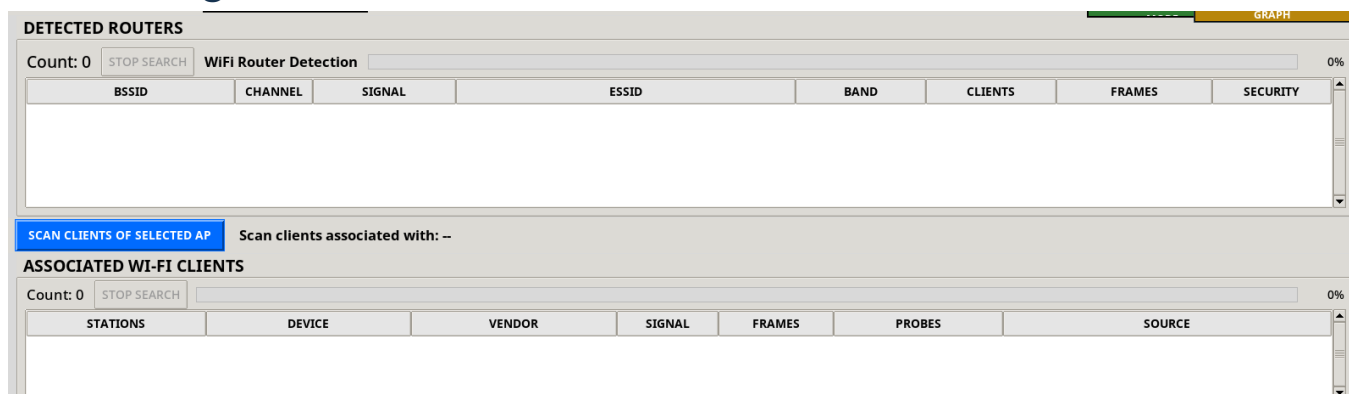


Figure 3 - Router detection and associated-client areas.

1. After monitor mode is active, press SCAN WI-FI NETWORKS.
2. Wait for routers to populate the DETECTED ROUTERS table.
3. Use BSSID and ESSID together to identify the intended access point. BAND and CHANNEL help distinguish 2.4 GHz and 5 GHz radios.
4. Select a router row before starting a client scan or passive capture.

The router table can display BSSID, CHANNEL, SIGNAL, ESSID, BAND, CLIENTS, FRAMES and SECURITY. The client count can be updated as the scan receives additional frames. Manual column sorting is intended for a paused or completed scan so live updates do not fight the user-defined order.

7. Scanning associated Wi-Fi clients

With an AP selected, press SCAN CLIENTS OF SELECTED AP. The client area shows STATIONS, DEVICE, VENDOR, SIGNAL, FRAMES, PROBES and SOURCE. Selecting a client also updates the CLIENT field in the lower capture/control area.

A MAC directly observed as a Wi-Fi station is treated as Wi-Fi evidence with priority over DS-side heuristics. This is an important protection against false LAN classifications.

8. LAN-side device analysis

LAN-SIDE DEVICES

Count: 0

VENDOR	ROLE	LIKELIHOOD	SC	MAC ADDRESS	SOURCE

BLOCK/CAPTURE 0%

BSSID: PASSIVE CAPTURE BSSID: --

CLIENT: START STOP DURATION (sec): 120 EXPO

CHANNEL:

BLOCK

ROUTER START STOP

CLIENT

DISRUPTION BSSID: -- / CLIENT: --

START STOP DURATION (sec): 5

☐ STOP WHEN HANDSHAKE IS FOUND PAUSE (sec): 3 EXPORT H.

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Figure 4 - LAN-side result table and lower capture controls.

The LAN-SIDE DEVICES table is designed for endpoints observed on the Distribution System side of the selected BSSID. Columns include VENDOR, ROLE, LIKELIHOOD, SC, MAC ADDRESSES and SOURCE. The short SC column is the numeric LAN score, while LIKELIHOOD converts the score into a readable level.

LAN score	Displayed likelihood
75-100	HIGH
45-74	MODERATE
0-44	LOW

8.1 Evidence used by the passive LAN classifier

- From-DS Address 3 source observations.
- To-DS Address 3 destination observations.
- WDS Address 3 / Address 4 evidence when present.
- Bidirectional DS evidence, duration and repeated observations.
- ARP / Ethernet evidence when decodable.
- Exclusion of MAC addresses directly observed as Wi-Fi stations.
- Additional ambiguity handling for related dual-band / mesh radios.

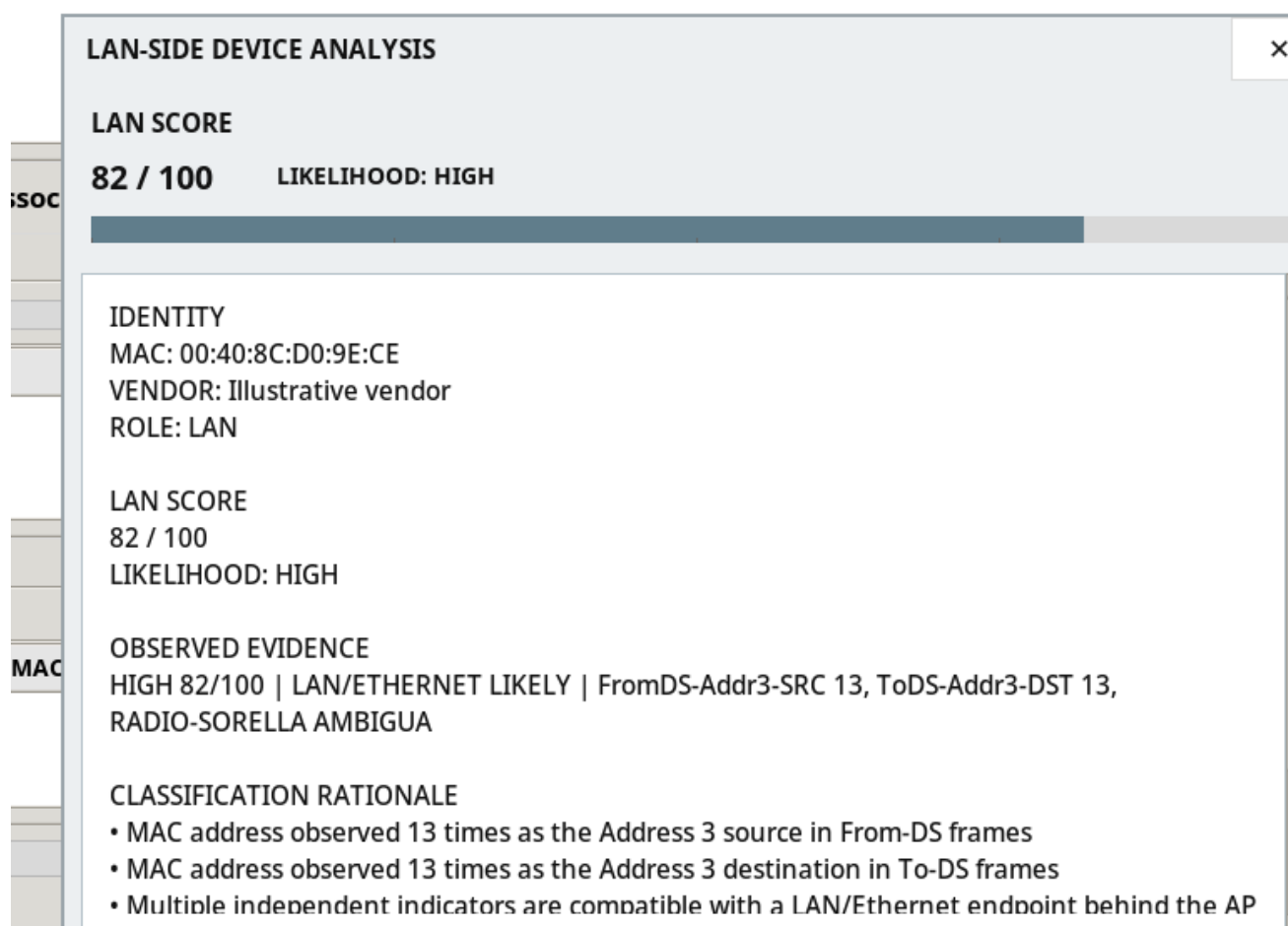


Figure 5 - Actual LAN detail window with illustrative data. The score bar, identity, evidence and classification rationale are generated by the program.

Clicking a LAN row opens the LAN-SIDE DEVICE ANALYSIS window. Use the evidence text as the main explanation of why the device was kept or rejected. Strong bidirectional evidence is more meaningful than a single Address 3 appearance.

9. Passive capture and capture controls

The PASSIVE CAPTURE panel follows the currently selected BSSID. The default duration shown by this build is 120 seconds. During a capture, progress is reflected in the BLOCK/CAPTURE progress bar and the analysis results are updated from the resulting PCAP.

1. Select the intended router in DETECTED ROUTERS.
2. Confirm that the BSSID and channel shown in the lower-left fields match the target.
3. Choose a capture duration when a value other than the default is required.
4. Press START under PASSIVE CAPTURE and allow the capture to complete, or press STOP when you intentionally want to finish early.
5. After analysis, review LAN-SIDE DEVICES, OBSERVED MAC ADDRESSES, POSSIBLE CAMERA DEVICES and HANDSHAKE together rather than relying on a single table.

9.1 Block / Disruption controls

The ROUTER and CLIENT block controls and the DISRUPTION panel are separate from passive capture. START begins the configured test action and STOP terminates it. The STOP WHEN HANDSHAKE IS FOUND option can stop the disruption workflow when a complete handshake condition is detected. Use these functions only in an authorized test environment.

Passive capture is the recommended default for observation and classification. Interference controls are not required to identify ordinary routers, clients, DS-side candidates or many camera-like traffic patterns.

10. Handshake panel

The screenshot shows a software interface with three main panels. The top-left panel, titled 'HANDSHAKE', contains four lines for M1, M2, M3, and M4, each followed by a double dash '--'. Below these is a large button labeled 'ABSENT'. To the left of this panel is a vertical slider with a '0%' mark at the bottom. The top-right panel, titled 'OBSERVED MAC ADDRESSES', is a table with columns: MAC, VENDOR, CLASS, EVIDENCE, NOTES, and SOURCE. The bottom panel, titled 'POSSIBLE CAMERA DEVICES', shows 'Numero: 0' and a table with columns: MAC, VENDOR, SCORE, LIKELIHOOD, TX/RX, SOURCE, and DURATION. At the bottom right of this panel are two buttons: 'DETTAGLI' and 'BLOCCA DISPOSITIVO SELEZIONATO'.

Figure 6 - Handshake, observed-MAC and camera-analysis areas.

The HANDSHAKE box shows M1, M2, M3 and M4 separately. The final state is displayed below the four message lines. A complete 4-way exchange can be latched so a later refresh does not immediately erase a previously confirmed result during the same analysis session.

Use the handshake display as an observation result. The presence of only some messages should be interpreted as partial evidence rather than a complete handshake.

11. Observed MAC addresses

OBSERVED MAC ADDRESSES is the broad classification table. It can contain the selected access point, associated Wi-Fi stations, LAN-side candidates, multicast/broadcast addresses and uncertain MAC addresses. The English columns are MAC, VENDOR, CLASS, EVIDENCE, NOTES and SOURCE.

Class	Meaning
ACCESS POINT	The selected or known infrastructure BSSID.
ASSOCIATED WI-FI CLIENT	A station directly observed on the radio link.
LAN-SIDE CANDIDATE	An endpoint with sufficient DS-side evidence.
OTHER / UNCERTAIN	Observed, but the role cannot be established with enough confidence.
MULTICAST / BROADCAST	Group/service address; not counted as a normal device.

12. Possible camera devices

POSSIBLE CAMERA DEVICES is a probabilistic traffic-analysis view. It does not decrypt payloads and does not require application-layer content. The build scores observable characteristics such as traffic direction, continuity, frame sizes, timing, radio stability and presence. A vendor/OUI match may contribute evidence but does not, by itself, prove that a device is a camera.

Camera score	Displayed level
80-100	HIGHLY LIKELY
62-79	LIKELY
42-61	POSSIBLE
0-41	LOW CONFIDENCE

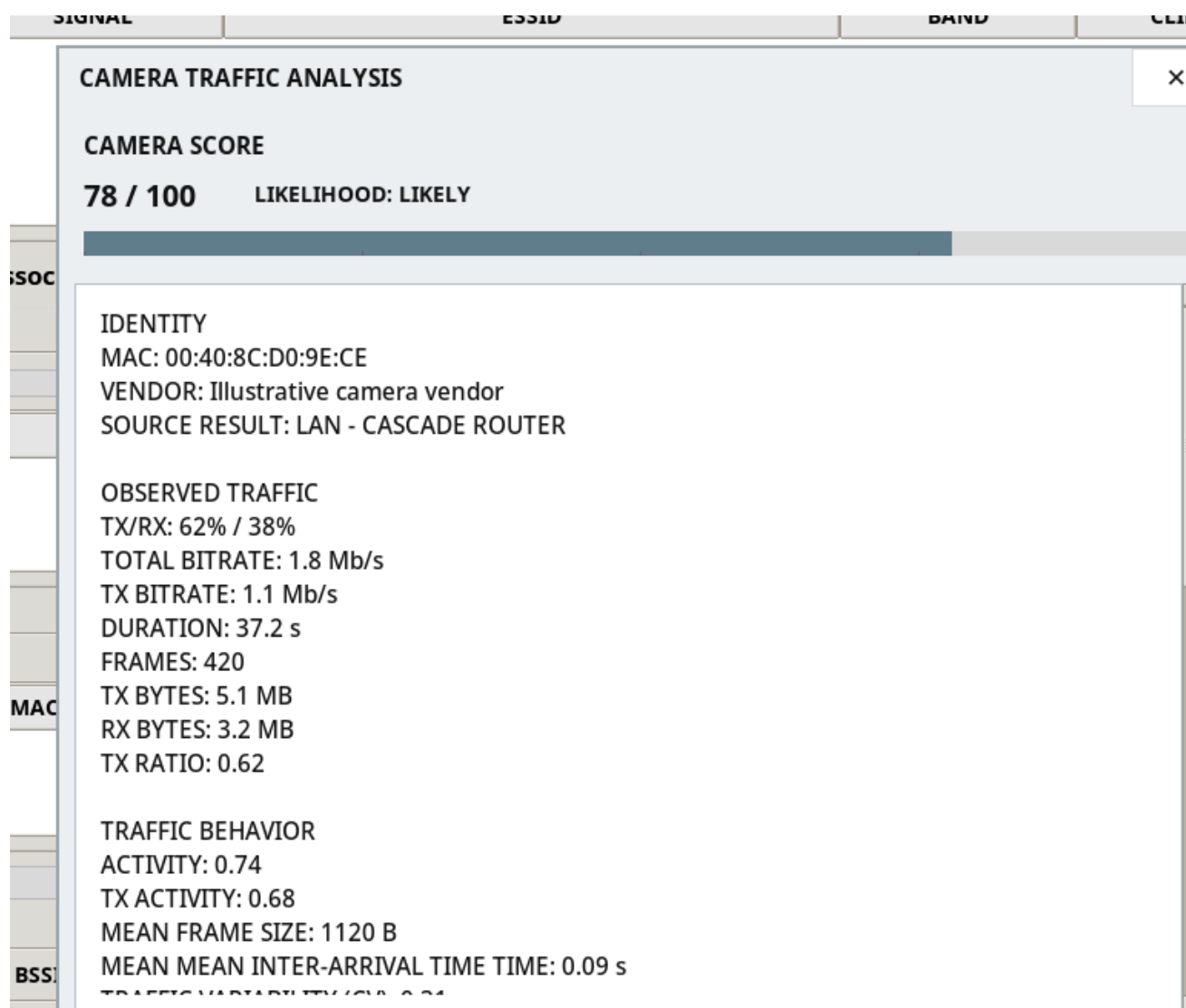


Figure 7 - Actual CAMERA TRAFFIC ANALYSIS window with illustrative values. Real score components depend on the captured traffic.

The camera detail window includes the source result, TX/RX balance, bitrate measurements, duration, frame count, activity metrics, retry ratio, RSSI, percentile rates, presence/continuity values and a score breakdown when those fields are available.

13. Dual-band and cascade-router logic

13.1 Related 2.4 GHz and 5 GHz radios

Modern routers often expose related BSSIDs for different frequency bands. A client on the sibling radio can appear in DS-side address fields of traffic observed on the selected radio. Earlier logic could therefore produce either false LAN entries or, after aggressive filtering, hide genuine LAN devices.

WIFI_438 keeps the conservative radio-first rule while avoiding a blanket rejection of DS endpoints:

- If a MAC is directly observed as a Wi-Fi station, that radio evidence wins and the MAC is not published as a LAN-side device.
- If a MAC is not observed as a station and has strong repeated DS evidence, it can remain a LAN-side candidate even when a sibling 2.4/5 GHz radio exists.
- Sibling-radio ambiguity lowers confidence instead of automatically deleting a strongly supported candidate.
- Weak or one-direction-only Address 3 evidence remains insufficient in an ambiguous dual-band/mesh topology.

13.2 Cascade-router workflow

The program can correlate a second router/BSSID with the first capture and propose a follow-up passive scan. Cascade-router prompts use a distinct visual treatment so they are not confused with same-device dual-band prompts. When a second passive scan is accepted, the program is designed to preserve the first-scan results and integrate the new source instead of replacing the earlier information.

When reviewing a cascade result, check SOURCE in the LAN and camera tables. Provenance is important because the same device may have been discovered in a first scan, a related-band scan, or a cascade-router follow-up scan.

14. Export graph, debug and files

14.1 Export Graph

EXPORT GRAPH builds a topology-oriented view from the program history. The application keeps a cumulative in-memory graph of routers, clients, LAN-side devices, cameras and relationships for the current program session. This makes it possible to represent information gathered across related scans instead of showing only the last table refresh.

14.2 Debug

DEBUG opens the command/result diagnostic panel. It is useful when a scan or external command does not behave as expected because the application logs invoked commands and selected results. Use it to distinguish a GUI problem from a missing executable, permission problem, adapter problem or command failure.

14.3 Capture and export locations

Location	Purpose
wifi_audit_gui	Working/output folder created next to the program file.
CATTURE	Archive folder for saved capture files.
EXPORT CAPTURE	Exports the selected/available capture through the program interface.
EXPORT HANDSHAKE	Exports handshake-related data when the corresponding result is available.

15. Night mode and screen controls

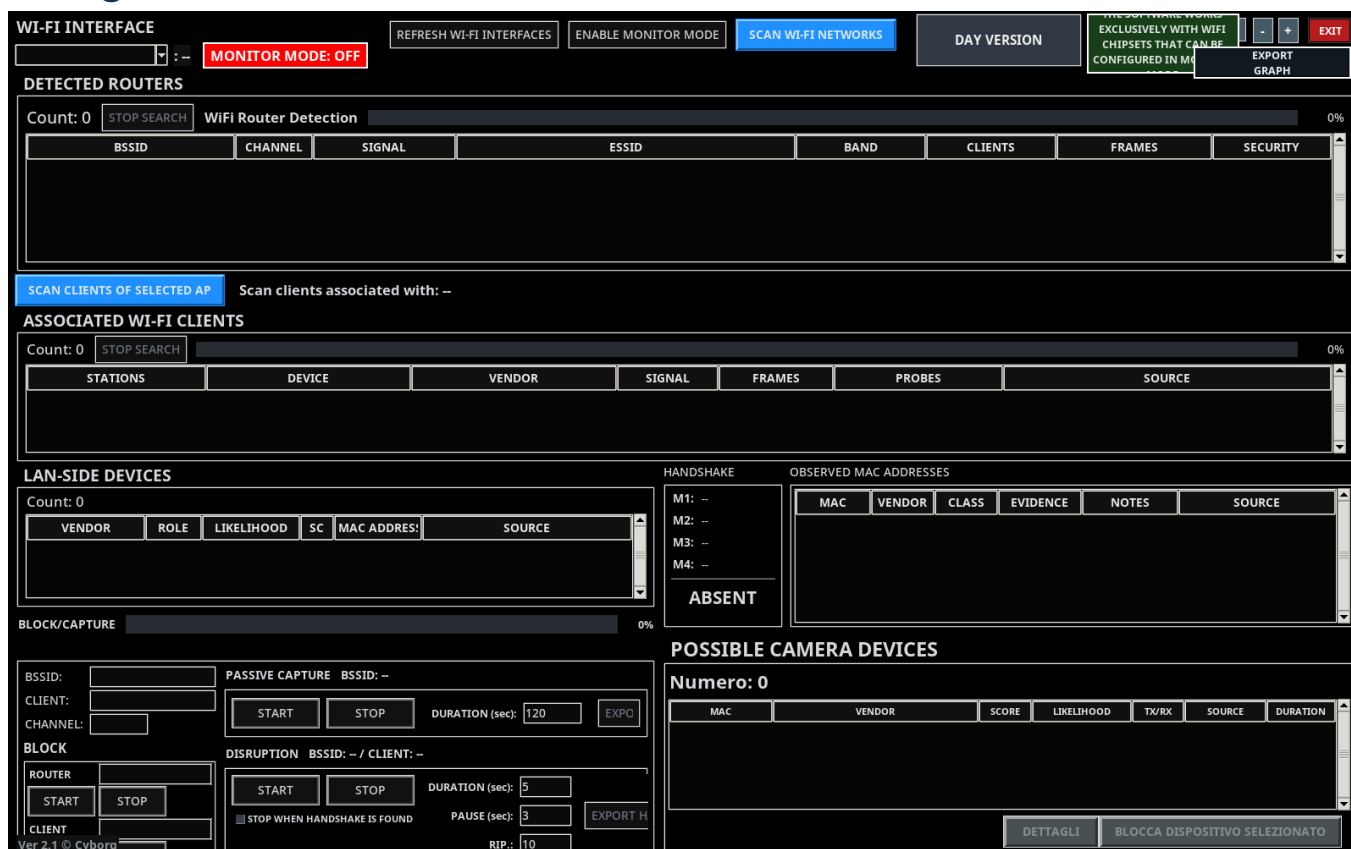


Figure 8 - Actual program in Night Version. The same tables and controls remain available with a dark palette.

NIGHT VERSION changes the interface palette while preserving the layout. When night mode is active, the button changes to DAY VERSION. The application also includes full-screen handling and vertical-size controls so the dense interface can be adapted to different displays.

- Use F11 to toggle full screen.
- Use Esc to leave full screen if required.
- Use + and - to adjust only the GUI height without changing the current width.
- Detail popups are created inside the application area so the desktop should not become the main focus while a result is being inspected.

16. Understanding scores and evidence

Scores are decision aids, not absolute proof. Always read the score together with the evidence text, source/provenance and whether the MAC has direct Wi-Fi station evidence.

Result	Use the score for	Do not assume
LAN score	Relative strength of DS-side endpoint evidence.	That the device is physically cabled just because the score is high.
Camera score	How compatible observed traffic is with the camera/always-on profile.	That the payload has been identified or decrypted.
Signal / RSSI	Radio strength/stability of directly observed wireless traffic.	Physical distance without accounting for antenna, walls and transmit power.
Handshake state	Whether the expected EAPOL messages were observed.	That missing messages do not exist outside the capture window.

16.1 Source / provenance

SOURCE is especially important in WIFI_438. It tells you where a row came from - for example the current BSSID, a related band, a cascade-router scan, or another analysis stage. When two scans are chained, provenance helps you keep the first and second observations separate while still viewing them in one workflow.

17. Recommended authorized workflow

1. Start the program with root privileges and refresh the Wi-Fi interface list.
2. Select the external USB Wi-Fi adapter and enable monitor mode.
3. Scan Wi-Fi networks and select the intended AP by checking BSSID, ESSID, channel and band.
4. Scan associated clients if client identity is relevant to the investigation.
5. Run a passive capture on the selected BSSID. Allow enough time for low-traffic DS-side devices to produce observable frames.
6. Review the general MAC table first, then the LAN-side table and the camera table. Open detail windows for candidates that matter.
7. If the software proposes a related-band or cascade-router passive scan, verify the old and new BSSID/SSID information before accepting it.
8. Use SOURCE to compare first-scan and follow-up observations. Do not treat a single Address 3 appearance as proof of a LAN device.
9. Export the graph or capture when you need a record of the current investigation.

For low-traffic devices, a longer passive observation can be more useful than repeatedly starting and stopping short scans. The classifier uses persistence and bidirectional evidence as important confidence signals.

18. Troubleshooting

Symptom	Checks
No Wi-Fi interfaces appear	Press REFRESH WI-FI INTERFACES. Confirm the USB adapter is present in the OS and visible to iw/sysfs.
Monitor mode will not enable	Run as root. Verify the adapter is external USB and that the PHY/driver declares monitor mode.
Router table stays empty	Confirm monitor mode is ON, the adapter supports the required band, and the scan has been allowed to run long enough.
Clients are missing	Select the correct BSSID and allow client traffic to occur. A quiet client may not be visible in a short window.
A real LAN device is missing	Check whether the MAC was directly observed as Wi-Fi. For quiet DS endpoints, collect a longer passive sample and inspect the evidence/detail view.
Too many LAN candidates	Check SOURCE and direct Wi-Fi evidence. Weak one-direction DS observations should remain uncertain; related radios can create ambiguous Address 3 traffic.
Camera not detected	Camera scoring is traffic-pattern based. An idle device or short capture may not provide enough activity for a high score.
Export or external tool fails	Open DEBUG and check whether the relevant executable exists and whether the command returned an error.
Desktop becomes visible around a popup	Use F11/full-screen mode and verify that the current build is the English WIFI_438 file documented here.

19. Quick reference

Control / area	Short description
REFRESH WI-FI INTERFACES	Refresh available wireless adapters.
ENABLE MONITOR MODE	Put the selected external USB adapter into monitor mode.
SCAN WI-FI NETWORKS	Discover nearby access points.
SCAN CLIENTS OF SELECTED AP	Search for stations associated with the selected BSSID.
PASSIVE CAPTURE - START / STOP	Capture traffic for analysis without using the disruption controls.
LAN-SIDE DEVICES	DS-side endpoint candidates with score, likelihood and source.
OBSERVED MAC ADDRESSES	General MAC classification and evidence table.
POSSIBLE CAMERA DEVICES	Probabilistic camera/always-on traffic candidates.
HANDSHAKE	M1-M4 observation and final handshake status.
DETAILS	Open the selected device analysis view when enabled.
EXPORT GRAPH	Export the cumulative topology view.
DEBUG	Open command/result diagnostics.
NIGHT VERSION / DAY VERSION	Switch theme.
EXIT	Close the application.

Version note

This help file is based on the English source file WIFI_438_EN_FIX_LAN_DUALBAND_CASCADE.py. The source still contains some legacy internal labels/version strings inherited from earlier builds; the operational behavior documented here follows the WIFI_438 English build file used to generate the screenshots.

End of user guide.